كلية التربية

الفرقة الاولي عام مميز

شعبة رياضيات

الفصل الدراسي الأول

2024-2023

تاريخ الامتحان: 2023/12/31

نموذج اجابة+ صورة من الاسئلة

ورقة كاملة

المادة: Calculus (1)

أستاذ المادة : د . أحمد مصطفي عبدالباقي مجاهد

أستاذ بقسم الرياضيات بكلية العلوم ببنها

صورة من الاسئلة

Benha University
Faculty of Education

Subject: Rigid body dynamics
Time: 1 hours
Class: 3st Year Students,
(25-12-2023)

Answer the following questions:

Q1. a) Determine whether the function is even, odd, or neither.

$$f(x) = x^4, \quad g(x) = x^{-1}$$

b) Use the definition of the derivatives to find the first derivative of $f(x) = \sqrt{x}$

Q2.

a) Evaluate the following limits $\lim_{n\to\infty} \frac{5x^2+8x+3}{3x^2+2}$, $\lim_{n\to-\infty} \frac{11x+2}{2x^3-1}$, $\lim_{x\to 0^+} (1 + sin4x)^{cotx}$

b) Find the horizontal and vertical asymptotes of the function $f(x) = \frac{-8}{x^2-4}$, **then sketch it**.

Q3. a) Differentiate the following functions:

$$f(x) = \frac{secx}{1 + tanx}, \quad y = cot^{-1}\left(\frac{1}{x^2}\right), \quad y = x^x + sinx^{cosx}$$

b) Find the equation of the tangent and normal lines to the graph $f(x) = \sqrt{x^2 + 3}$ at the point (-1, 2).

Q4. a) Find the Maclaurin series for the following function $f(x) = ln(1 + x)$.

b) Find the points of inflection and the intervals on which $f(x) = 3x^5 - 5x^4 + 1$ is concave up and concave down.

Model Answer

Q1

a)

(a) $f(-x) = (-x)^4 = x^4$. Thus, $f(x) = f(-x)$, and f is even.

(b) $g(-x) = (-x)^{-1} = -x^{-1}$. Thus, $g(-x) = -g(x)$, and g is odd.

b)

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} \\
&= \lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} \frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}} \\
&= \lim_{h\to 0} \frac{x+h-x}{h(\sqrt{x+h}+\sqrt{x})} \\
&= \lim_{h\to 0} \frac{1}{(\sqrt{x+h}+\sqrt{x})} \\
&= \frac{1}{2\sqrt{x}}
\end{aligned}$$

Q2

a)

$$\begin{aligned}
\lim_{x\to\infty} \frac{5x^2+8x-3}{3x^2+2} &= \lim_{x\to\infty} \frac{5+(8/x)-(3/x^2)}{3+(2/x^2)} \\
&= \frac{5+0-0}{3+0} = \frac{5}{3}
\end{aligned}$$

where we divide numerator and denominator by x^2.

$$\begin{aligned}
\lim_{x\to-\infty} \frac{11x+2}{2x^3-1} &= \lim_{x\to-\infty} \frac{(11/x^2)+(2/x^3)}{2-(1/x^3)} \\
&= \frac{0+0}{2-0} = 0
\end{aligned}$$

where we divide numerator and denominator by x^3.

Since $\lim_{x\to 0^+}(1+\sin 4x)^{\cot x} = 1^\infty$ *(indeterminate power), putting*

$$y = (1+\sin 4x)^{\cot x} \Rightarrow \lim_{x\to 0^+} y?$$

Taking the natural logarthim ln, *we have*

$$\ln y = \cot \ln(1+\sin 4x) = \frac{\ln(1+\sin 4x)}{\tan x}$$
$$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} \frac{\ln(1+\sin 4x)}{\tan x} = \frac{0}{0}$$

Now, we can use L'Hospital rules

$$\begin{aligned}\ln \lim_{x\to 0^+} y &= \lim_{x\to 0^+} \frac{4\cos 4x/(1+\sin 4x)}{\sec^2 x} \\ &= \lim_{x\to 0^+} \frac{4\cos 4x}{\sec^2 x(1+\sin 4x)} \\ &= \lim_{x\to 0^+} \frac{4\cos 4x \cos^2 x}{(1+\sin 4x)} = 1\end{aligned}$$

Hence, we obtain

$$\lim_{x\to 0^+} y = e$$

b)

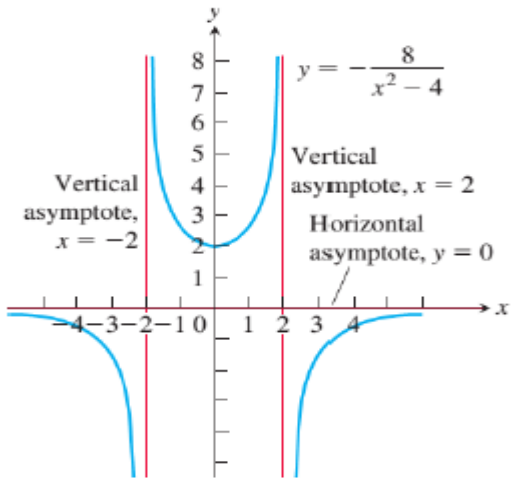
Solution: We are interested in the behavior as $x \to \pm\infty$ and as $x \to \pm 2$, where the denominator is zero. Notice that f is an even function of x, so its graph is symmetric with respect to the y-axis.

(a) The behavior as $x \to \pm\infty$. Since $\lim_{x\to\infty} f(x) = 0$, the line $y = 0$ is a horizontal asymptote of the graph to the right. By symmetry it is an asymptote to the left as well (Figure 2.22). Notice that the curve approaches the x-axis from only the negative side (or from below). Also, $f(0) = 2$.

The behavior as $x \to \pm 2$. Since

$$\lim_{x\to 2^+} f(x) = -\infty \quad \text{and} \quad \lim_{x\to 2^-} f(x) = \infty$$

the line $x = 2$ is a vertical asymptote both from the right and from the left. By symmetry, the line $x = -2$ is also a vertical asymptote. There are no other asymptotes because f has a finite limit at all other points.



Q3

$$
\begin{aligned}
f'(x) &= \frac{(1+\tan x)\frac{d}{dx}(\sec x) - \sec x \frac{d}{dx}(1+\tan x)}{(1+\tan x)^2} \\
&= \frac{(1+\tan x)\sec x \tan x - \sec x \cdot \sec^2 x}{(1+\tan x)^2} \\
&= \frac{\sec x\,(\tan x + \tan^2 x - \sec^2 x)}{(1+\tan x)^2} \\
&= \frac{\sec x(\tan x - 1)}{(1+\tan x)^2}
\end{aligned}
$$

$$
\begin{aligned}
y' &= \left(\cot^{-1}\frac{1}{x^2}\right)' \\
&= -\frac{1}{1+\left(\frac{1}{x^2}\right)^2}\cdot\left(\frac{1}{x^2}\right)' \\
&= -\frac{1}{1+\frac{1}{x^4}}\cdot\left(-2x^{-3}\right) \\
&= \frac{2x^4}{(x^4+1)\,x^3} \\
&= \frac{2x}{1+x^4}
\end{aligned}
$$

Solution: *Consider*

$$y = y_1 + y_2$$

where $y_1 = x^x, \quad y_2 = (\sin x)^{\cos x}$ *Since*

$$\begin{aligned}
\ln y_1 &= \ln x^x \\
&= x \ln x \\
\frac{1}{y_1} y_1' &= x \frac{1}{x} + \ln(x) \\
y_1' &= x^x [1 + \ln(x)]
\end{aligned}$$

and

$$\begin{aligned}
\ln y_2 &= \ln(\sin x)^{\cos x} \\
&= (\cos x) \ln(\sin x) \\
\frac{1}{y_2} y_2' &= \frac{\cos^2 x}{\sin x} - \sin x \ln(\sin x) \\
y_2' &= (\sin x)^{\cos x} [\cot x \csc x - \sin x \ln(\sin x)]
\end{aligned}$$

Then

$$\begin{aligned}
y' &= y_1' + y_2' \\
&= x^x [1 + \ln(x)] + (\sin x)^{\cos x} [\cot x \csc x - \sin x \ln(\sin x)]
\end{aligned}$$

b)

Solution:

$$f(x) = \left(x^2+3\right)^{1/2}$$
$$f'(x) = \frac{1}{2}\left(x^2+3\right)^{-1/2} \cdot (2x)$$
$$f'(x) = \frac{x}{\sqrt{x^2+3}}$$

At the point $(-1,2)$, $f'(-1) = -1/2$ *and the equation of the line is*

$$y - y_1 = m\,(x - x_1)$$
$$y - 2 = -\frac{1}{2}(x+1)$$
$$2y - 4 = -x - 1$$
$$x + 2y = 3$$

Similarly, the equation of the normal line at the point $(-1,2)$ *is*

$$\frac{y-2}{x+1} = 2$$
$$y - 2 = 2(x+1)$$
$$y - 2x - 4 = 0$$

Q4

Solution

$$f(x) = \ln(1+x) \;\Rightarrow\; f(0) = \ln(1) = 0$$
$$f'(x) = \frac{1}{1+x} = (1+x)^{-1} \Rightarrow\; f'(0) = 1$$
$$f''(x) = \frac{-1}{(1+x)^2} \;\Rightarrow\; f''(0) = -1$$
$$f'''(x) = \frac{2}{(1+x)^3} \;\Rightarrow\; f'''^{(0)} = 2$$
$$f^{(4)}(x) = \frac{-6}{(1+x)^4} \;\Rightarrow\; f^{(4)}(0) = -6, \ldots\ldots\ldots$$

Substituting in the Maclaurin series we get

$$\therefore \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

b)

Solution: *The first derivative is* $f'(x) = 15x^4 - 20x^3$ *and*

$$f''(x) = 60x^3 - 60x^2 = 60x^2(x-1)$$

The zeros of $f''(x) = 60x^2(x-1)$ *are* $x = 0$ *and* $x = 1$. *They divide the* x*-axis into three intervals:* $(-\infty, 0)$, $(0, 1)$, *and* $(1, \infty)$. *We determine the sign of* $f''(x)$ *and the concavity of* f *by computing "test values" within each interval (Figure 7):*

Interval	*Test Value*	*Sign of* $f''(x)$	*Behavior of* $f(x)$
$(-\infty, 0)$	$f''(-1) = -120$	$-$	*Concave down*
$(0, 1)$	$f''\left(\frac{1}{2}\right) = -\frac{15}{2}$	$-$	*Concave down*
$(1, \infty)$	$f''(2) = 240$	$+$	*Concave up*

We can read off the points of inflection from this table:

- $c = 0$: *no point of inflection, because* $f''(x)$ *does not change sign at* 0.
- $c = 1$: *point of inflection, because* $f''(x)$ *changes sign at* 1 .

Dr. Ahmed Mostafa Megahed